BASIC PROPERTIES OF THE NORMALIZED BLOWUP OF A NAGATA SCHEME AT A POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: properness of the blowing up, preservation of the Nagata property, and finiteness of the normalization step.

Editorial note. The label lemma-Nagata-normalized-blowup names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-Nagata-normalized-blowup –start 1244 –end 1266 –local-id 28116 returned RESULT=OK (L28116 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Projectivity of blowing up for locally Noetherian schemes. Divisors, the lemma that blowing up is projective in the locally Noetherian case. Gives properness of the blowing up morphism.
- Nagata property is preserved under finite type morphisms. Morphisms, the lemma on the Nagata property for schemes of finite type over a Nagata base. Shows that the blowup is Nagata.
- Finiteness of normalization for Nagata schemes. Morphisms, the lemma that the normalization of a Nagata scheme is finite, and the lemma that the normalization is normal. Gives finiteness and normality of the normalization step, hence properness of the composite.

Source reference. Original reference 0BBS: sources/stacks-resolve/pinned-original-source.tex, source lines 1222–1222. Prerequisite pointer invoked at proof line(s) 1246; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02NS: sources/stacks-chow/tags.txt, source lines 3491–3491. Prerequisite pointer invoked at proof line(s) 1254; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Prerequisite pointer invoked at proof line(s) 1261; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Prerequisite pointer invoked at proof line(s) 1260; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 035A: sources/stacks-chow/tags.txt, source lines 4121–4121. Prerequisite pointer invoked at proof line(s) 1256; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 035S: sources/stacks-chow/tags.txt, source lines 4139–4139. Prerequisite pointer invoked at proof line(s) 1258; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 035Q: sources/stacks-chow/tags.txt, source lines 4137–4137. Prerequisite pointer invoked at proof line(s) 1264; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Definition 0BBS if X is Nagata, then the normalized blowing up of X at x is normal, Nagata, and proper over X.*

Proof. The blowup morphism $X' \to X$ is proper (as X is locally Noetherian we may apply Divisors, Lemma 02NS). Thus X' is Nagata (Morphisms, Lemma 035A). Therefore the normalization $X'' \to X'$ is finite (Morphisms, Lemma 035S) and we conclude that $X'' \to X$ is proper as well (Morphisms, Lemmas 01WN and 01W3). It follows that the normalized blowing up is a normal (Morphisms, Lemma 035Q) Nagata algebraic space. □